SATHSARA RANASOORIYA

FULL STACK AI ENGINEER INTERN

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https://github.com/Sathsararns | https://sathsararns.github.io/portfolio

PROFESSIONAL SUMMARY

Early-career developer experienced in building full-stack AI applications using Python, FastAPI, React, Next.js, and TypeScript. Skilled in LLM integration, Retrieval-Augmented Generation (RAG), LangGraph agent workflows, REST APIs, and machine learning pipelines. Strong problem-solving and collaborative development skills, seeking a Full Stack AI Engineer Internship.

EDUCATION

Bachelor of Science (BSc) in Physical Science, University of Ruhuna Aug 2023 - Sep 2026

- **Main Modules**: Pure Mathematics, Applied Mathematics, and Computer Science.

PROFESSIONAL EXPERIENCE

Business Analyst Intern (Part time) | H M R D Cooperate (Pvt) Ltd Jan 2026 - Jul 2026

- Gathered and analyzed stakeholder and user requirements to support software solution planning and development.
- Translated business requirements into functional documentation to support developer understanding and implementation.
- Collaborated with stakeholders and development teams to clarify requirements and align software solutions with business objectives.

TECHNICAL SKILLS

- **Languages**: Python, TypeScript, JavaScript, SQL, Java.
- **Frontend**: React.js, Next.js, Tailwind CSS, HTML/CSS.
- **Backend**: FastAPI, Node.js, Express.js, REST APIs, Pydantic, SQLAlchemy.
- **Generative AI**: LLM Integration, Prompt Engineering, RAG, LangGraph, Agentic AI, Pinecone.
- **Machine Learning & MLOps**: scikit-learn, XGBoost, PySpark, Pandas, NumPy, MLflow.
- **Databases & Data Engineering**: PostgreSQL, MongoDB, Redis, Apache Kafka, Airflow.
- **DevOps & Tools**: Docker, Git, GitHub, Postman.

SOFT SKILLS

- Cross-functional collaboration and stakeholder communication.
- Analytical thinking, problem-solving, and debugging.
- Requirements analysis and technical documentation.
- Adaptability, continuous learning, and attention to detail.

ENGINEERING EXPERIENCE (PROJECTS)

Voya AI – Full-Stack RAG Travel Planning Assistant GitHub Link

- **Tech Stack**: React, TypeScript, FastAPI, Groq LLM, Pinecone, PostgreSQL, Redis, Docker
- Developed a full-stack AI travel planner using React, TypeScript, and FastAPI, featuring authentication, persistent conversations, and personalized itineraries
- Implemented Retrieval-Augmented Generation (RAG) using Pinecone vector search and Groq-hosted LLMs to generate context-aware travel recommendations.
- Built context-aware AI workflows with conversation memory, structured LLM outputs, rule-based response validation, and Server-Sent Events (SSE).

WedaHub – AI-Powered Service Booking Platform (Team Project) GitHub Link

- **Tech Stack**: React, Node.js, Express.js, MongoDB, FastAPI, LangGraph, Groq LLM, Socket.IO
- Contributed to a full-stack service booking platform featuring provider discovery, booking management, dashboards, authentication, and reviews.
- Integrated a LangGraph-based AI assistant with Groq LLMs for structured requirement extraction, provider recommendations, and multi-turn booking workflows.
- Connected React, Express.js, and FastAPI through REST APIs, with MongoDB persistence and Socket.IO-based booking status updates.

Opsyntra – Full-Stack Churn Prediction & Streaming Analytics GitHub Link

- **Tech Stack**: Next.js, TypeScript, FastAPI, scikit-learn, PySpark, Kafka, PostgreSQL, MLflow, Airflow, Docker
- Developed a Next.js and FastAPI application supporting individual and batch customer churn predictions, model evaluation, and analytics dashboards.
- Built scikit-learn and PySpark ML pipelines with MLflow tracking; the selected Random Forest achieved **0.844 ROC-AUC and 79.7% recall** on a held-out test dataset.
- Implemented Kafka-based event processing, PostgreSQL analytics, and replay handling, with Airflow workflows and Docker Compose for local orchestration.

ACHIEVEMENTS

- **1st Runner-Up**: **CodeRally VII (Intermediate Tier)** | Informatics Institute of Technology (IIT). 2026
- **1st Runner-Up**: **Road to Rush Coder 3.0** | University of Ruhuna. 2025

CERTIFICATIONS

IBM AI Engineering | Coursera 2026
Machine Learning and Deep Learning for Software Engineers | Coursera 2026

CORE COMPETENCIES

- Full-Stack AI Application Development and REST API Integration.
- LLM Integration, Retrieval-Augmented Generation (RAG), and Vector Search.
- Agentic AI Workflow Development and Orchestration.
- Machine Learning Pipeline Development, Model Evaluation, and MLOps.
- Event-Driven Data Processing, Streaming Analytics, and Database Integration.

REFERENCES

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